Huawei Access Network Optical Transport Failure Resolution for Node ID: WHI-ACC-50

Node: WHI-ACC-50 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_9358

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) within the Access role of the Huawei equipment at the Whitefield East POP, denoted by Node ID WHI-ACC-50, is precipitated by a concatenation of factors intrinsic to the optical transport layer. Specifically, this failure domain is occasioned by the susceptibility of the SFP/XFP modules to degradation, which can occur due to a plethora of reasons including, but not limited to, excessive temperature fluctuations, humidity, and the inherent susceptibility of these modules to degradation over time. Furthermore, fiber cuts, which can result from either accidental damage during maintenance activities or intentional severance, can also precipitate this failure. The Rx/Tx power loss, a critical metric in the optical domain, can also lead to this failure when the power levels exceed the stipulated thresholds, thereby disrupting the optical signal transmission. The Huawei equipment, operating with its proprietary CLI dialect, necessitates a nuanced understanding of its diagnostic and resolution commands to effectively troubleshoot and rectify this failure. The severity of this failure, categorized as Major (ALARM_9358), underscores the criticality of promptly addressing this issue to prevent protracted service outages and potential revenue loss.

Pre-requisite Checks

- Dispatch remote hands to Whitefield East POP for on-site inspection and intervention.
- Verify the status of redundant links to ensure that traffic is properly rerouted.
- Check the environmental conditions (temperature, humidity) at the site.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

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Command: display interface, Expected Output: Interface status and statistics, Description: This command isolates issues in the data plane by displaying the current status and statistics of all interfaces, thereby helping to identify any anomalies or errors that may be contributing to the Optical Transport Failure.
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Command: display optical-module, Expected Output: Optical module information, Description: This command provides detailed information about the optical modules, including their type, temperature, and voltage, which can help in diagnosing issues related to SFP/XFP degradation.

Command: display alarm, Expected Output: Current alarms, Description: This command displays all current alarms on the system, including the ALARM_9358 related to Optical Transport Failure, thereby facilitating the identification of the root cause of the failure.
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Resolution Steps

Step 1: Step: Replace Degraded SFP/XFP Module, Command: replace optical-module, Description: This step involves physically replacing the degraded SFP/XFP module with a new one to restore the optical signal transmission. The 'replace optical-module' command is used to logically replace the module in the system.

Step 2: Step: Repair or Replace Damaged Fiber, Command: no shutdown interface, Description: If the fiber cut is identified as the root cause, this step involves repairing or replacing the damaged fiber to reestablish the optical connection. Once the physical repair is done, the 'no shutdown interface' command is used to bring the interface back up.

Step 3: Step: Adjust Rx/Tx Power Levels, Command: set optical-module tx-power, Description: This step involves adjusting the transmit power levels of the optical module to ensure they are within the stipulated range. The 'set optical-module tx-power' command is used to adjust the power levels, thereby preventing Rx/Tx power loss.

Post-Check Verification

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Command: display interface, Expected Output: Interface status and statistics indicating no errors, Description: After implementing the resolution steps, this command is used to verify that the interface is functioning correctly and that there are no errors or alarms indicating a continued failure.

Command: display optical-module, Expected Output: Optical module information indicating normal operation, Description: This command is used to verify that the optical module is operating within normal parameters, including temperature and voltage, after replacement or adjustment.

Command: display alarm, Expected Output: No ALARM_9358 or related alarms, Description: This command is used to verify that the Optical Transport Failure alarm (ALARM_9358) has been cleared, indicating a successful resolution of the issue.
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Roll-back Procedures

- Step: Revert to Original SFP/XFP Module, Command: replace optical-module original, Description: If the replacement of the SFP/XFP module does not resolve the issue, this step involves reverting back to the original module to rule out any issues with the new module. The 'replace optical-module original' command is used to logically revert the module in the system.

- Step: Restore Original Fiber Configuration, Command: shutdown interface, Description: If the repair or replacement of the fiber does not resolve the issue, this step involves restoring the original fiber configuration. The 'shutdown interface' command is used to bring the interface down before physically restoring the original fiber configuration.

- Step: Revert Rx/Tx Power Levels, Command: default optical-module tx-power, Description: If adjusting the Rx/Tx power levels does not resolve the issue, this step involves reverting the power

levels back to their default settings. The 'default optical-module tx-power' command is used to reset the power levels to their original values.